1. จุติณัฏฐ์ รัตนะมงคลกุล 6609650228
2. รพินทร์ นะราช 6609650624

Wireshark Lab 02 -- DNS

Assignment

1. Run nslookup to obtain the IP address of the web server for the Indian Institute of Technology in Bombay, India: www.iitb.ac.in. What is the IP address of www.iitb.ac.in

ANS

ใช้ nslookup หา IP Address ของ www.iitb.ac.in ได้ 103.21.124.133

```
Microsoft Windows [Version 10.0.26100.3476]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user>nslookup www.iitb.ac.in
Server:  UnKnown
Address:  192.168.0.1

Non-authoritative answer:
Name:    www.iitb.ac.in
Address:  103.21.124.133
```

2. What is the IP address of the DNS server that provided the answer to your nslookup command in question 1 above

ANS

```
Microsoft Windows [Version 10.0.26100.3476]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user>nslookup www.iitb.ac.in
Server:  UnKnown
Address:  192.168.0.1

Non-authoritative answer:
Name:    www.iitb.ac.in
Address:  103.21.124.133
```

IP Address ของ DNS คือ

Address: 192.168.0.1

Server: UnKnown

3. Did the answer to your nslookup command in question 1 above come from an authoritative or non-authoritative server?

ANS

```
C:\Users\user>nslookup www.iitb.ac.in
Server:  UnKnown
Address:  192.168.0.1

Non-authoritative answer:
Name:    www.iitb.ac.in
Address:  103.21.124.133
```

เป็น Non-authoritative answer มาจาก DNS Cache ของ server ที่ให้บริการ

4. Use the nslookup command to determine the name of the authoritative name server for the iit.ac.in domain. What is that name? (If there are more than one authoritative servers, what is the name of the first authoritative server returned by nslookup)? If you had to find the IP address of that authoritative name server, how would you do so?

ANS

```
C:\Users\user>nslookup -type=NS iitb.ac.in
Server:  UnKnown
Address:  192.168.0.1

Non-authoritative answer:
iitb.ac.in      nameserver = dns1.iitb.ac.in
iitb.ac.in      nameserver = dns2.iitb.ac.in
iitb.ac.in      nameserver = dns3.iitb.ac.in
```

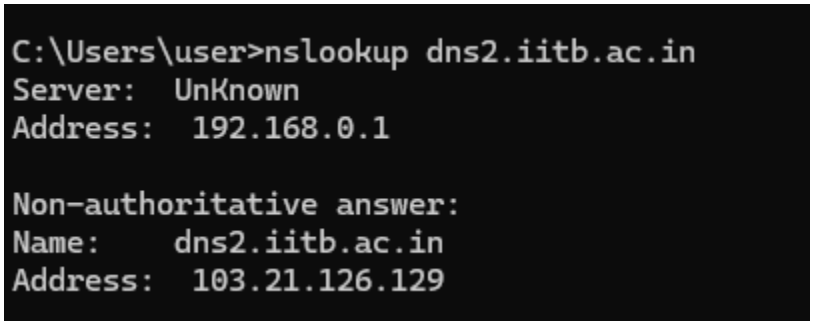
1) ชื่อของ authoritative name อันแรก คือ dns1.iitb.ac.in และ IP Address ที่ได้คือ 103.21.125.129

```
C:\Users\user>nslookup dns1.iitb.ac.in
Server:  UnKnown
Address:  192.168.0.1

Non-authoritative answer:
Name:    dns1.iitb.ac.in
Address:  103.21.125.129
```

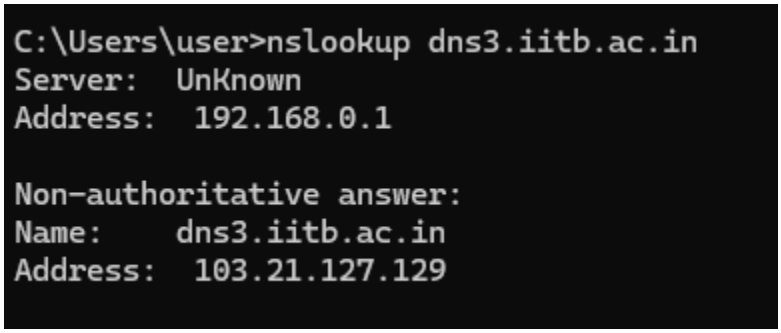
2) ชื่อของ authoritative name อันที่ 2 คือ dns2.iitb.ac.in

และ IP Address ที่ได้คือ 103.21.126.129



3) ชื่อของ authoritative name อันที่ 2 คือ dns3.iitb.ac.in

และ IP Address ที่ได้คือ 103.21.127.129

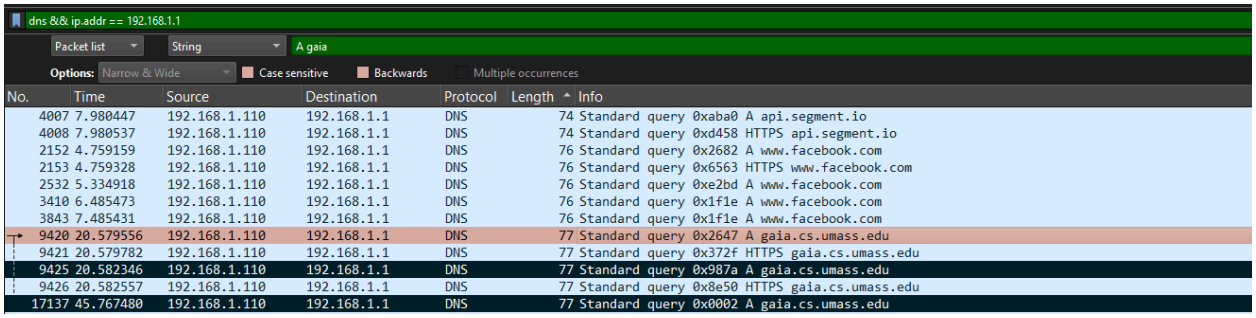


5. Locate the first DNS query message resolving the name gaia.cs.umass.edu. What

is the packet number6 in the trace for the DNS query message? Is this query

message sent over UDP or TCP?

ANS



มีแต่ข้อความ DNS Response และไม่มีข้อความ DNS Query สำหรับ gaia.cs.umass.edu

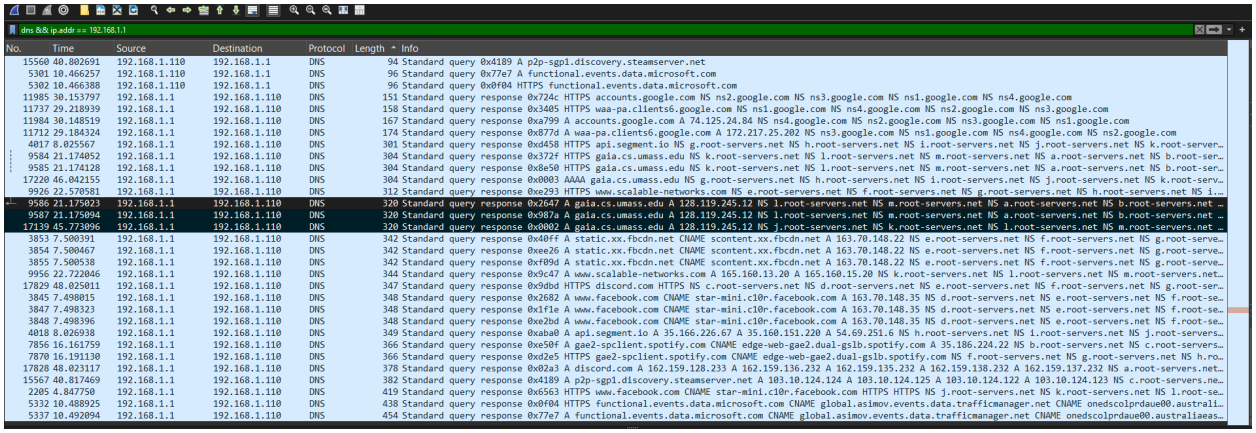
โดยทั่วไป DNS Query จะถูกส่งผ่าน UDP เว้นแต่มีขนาดใหญ่เกินจะส่งเป็น TCP

6. Now locate the corresponding DNS response to the initial DNS query. What is the

packet number in the trace for the DNS response message? Is this response

message received via UDP or TCP?

ANS



เลขแพ็กเก็ตของ DNS Query ไปยัง gaia.cs.umass.edu คือ No. 2590, 2631, 2687, 2908, 11189

ส่งผ่าน UDP เว้นแต่มีขนาดใหญ่เกินจะส่งเป็น TCP

แพ็กเก็ตแรกสุด: No. 2590

7. What is the destination port for the DNS query message? What is the source

port

of the DNS response message?

ANS

```
Frame 2590: 107 bytes on wire (856 bits), 107 bytes captured (856 bits) on interface \Device\NPF_{F9BE52F9-53E4-4BF8-84A4-6022B428AC8D}, id 0
Ethernet II, Src: TpLinkTechno_dd:83:16 (b0:be:76:dd:83:16), Dst: ASUSTekCOMPU_7f:44:73 (58:11:22:7f:44:73)
Internet Protocol Version 4, Src: 192.168.0.1, Dst: 192.168.0.104
Transmission Control Protocol, Src Port: 53, Dst Port: 56836, Seq: 1, Ack: 38, Len: 53
    Source Port: 53
    Destination Port: 56836
    [Stream index: 18]
    [Stream Packet Number: 8]
  ▸ [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 53]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 2055095054
    [Next Sequence Number: 54    (relative sequence number)]
    Acknowledgment Number: 38    (relative ack number)
    Acknowledgment number (raw): 182986446
    0101 .... = Header Length: 20 bytes (5)
  ▸ Flags: 0x018 (PSH, ACK)
    Window: 1054
    [Calculated window size: 269824]
    [Window size scaling factor: 256]
    Checksum: 0x1fc7 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  ▸ [Timestamps]
```

DNS Query(ปลายทาง): Destination Port 53

DNS Response(ต้นทาง): Source Port 53

Ex. No. 9586 : No. 9420

```
  9586 21.175023   192.168.1.1     192.168.1.110   DNS   320 Standard query response 0x2647 A gaia.cs.uma
  9587 21.175094   192.168.1.1     192.168.1.110   DNS   320 Standard query response 0x987a A gaia.cs.uma
 17139 45.773096   192.168.1.1     192.168.1.110   DNS   320 Standard query response 0x0002 A gaia.cs.uma
  3853 7.500391    192.168.1.1     192.168.1.110   DNS   342 Standard query response 0x40ff A static.xx.f

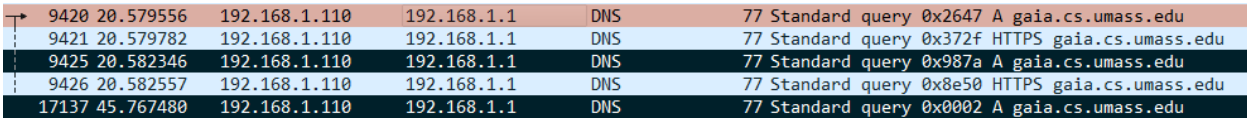
Frame 9586: 320 bytes on wire (2560 bits), 320 bytes captured (2560 bits) on interface \Device\NPF_{B2E73589-E582-475
Ethernet II, Src: TPLink_98:71:48 (5c:62:8b:98:71:48), Dst: GigaByteTech_a1:76:0b (d8:5e:d3:a1:76:0b)
Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.110
User Datagram Protocol, Src Port: 53, Dst Port: 51435
    Source Port: 53
    Destination Port: 51435
    Length: 286
    Checksum: 0x2830 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 31]
    [Stream Packet Number: 2]
  ▸ [Timestamps]
    UDP payload (278 bytes)
Domain Name System (response)
```

```
  9420 20.579556   192.168.1.110   192.168.1.1     DNS   77 Standard query 0x2647 A gaia.cs.umass.edu
  9421 20.579782   192.168.1.110   192.168.1.1     DNS   77 Standard query 0x372f HTTPS gaia.cs.umass.edu
  9425 20.582346   192.168.1.110   192.168.1.1     DNS   77 Standard query 0x987a A gaia.cs.umass.edu
  9426 20.582557   192.168.1.110   192.168.1.1     DNS   77 Standard query 0x8e50 HTTPS gaia.cs.umass.edu
 17137 45.767480   192.168.1.110   192.168.1.1     DNS   77 Standard query 0x0002 A gaia.cs.umass.edu

Frame 9420: 77 bytes on wire (616 bits), 77 bytes captured (616 bits) on interface \Device\NPF_{B2E73589-E582-475A-9C
Ethernet II, Src: GigaByteTech_a1:76:0b (d8:5e:d3:a1:76:0b), Dst: TPLink_98:71:48 (5c:62:8b:98:71:48)
Internet Protocol Version 4, Src: 192.168.1.110, Dst: 192.168.1.1
User Datagram Protocol, Src Port: 51435, Dst Port: 53
    Source Port: 51435
    Destination Port: 53
    Length: 43
    Checksum: 0x83fc [unverified]
    [Checksum Status: Unverified]
    [Stream index: 31]
    [Stream Packet Number: 1]
  ▸ [Timestamps]
    UDP payload (35 bytes)
Domain Name System (query)
```

8. To what IP address is the DNS query message sent?

ANS



9420	20.579556	192.168.1.110	192.168.1.1	DNS	77 Standard query 0x2647 A gaia.cs.umass.edu
9421	20.579782	192.168.1.110	192.168.1.1	DNS	77 Standard query 0x372f HTTPS gaia.cs.umass.edu
9425	20.582346	192.168.1.110	192.168.1.1	DNS	77 Standard query 0x987a A gaia.cs.umass.edu
9426	20.582557	192.168.1.110	192.168.1.1	DNS	77 Standard query 0x8e50 HTTPS gaia.cs.umass.edu
17137	45.767480	192.168.1.110	192.168.1.1	DNS	77 Standard query 0x0002 A gaia.cs.umass.edu

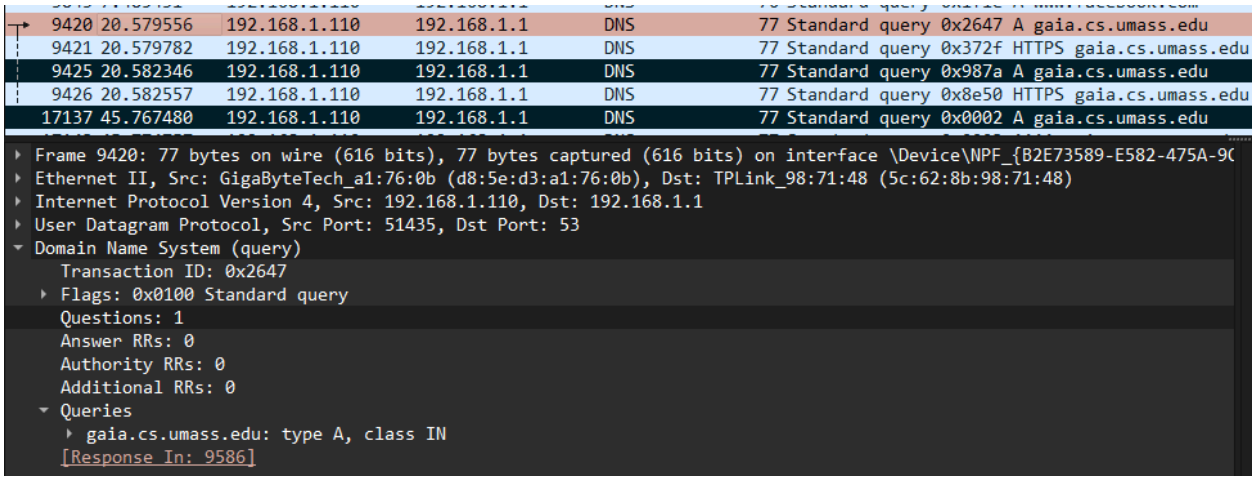
คำขอ DNS ถูกส่งไปยัง IP Address 192.168.1.110

9. Examine the DNS query message. How many “questions” does this DNS message

contain? How many “answers” answers does it contain?

ANS

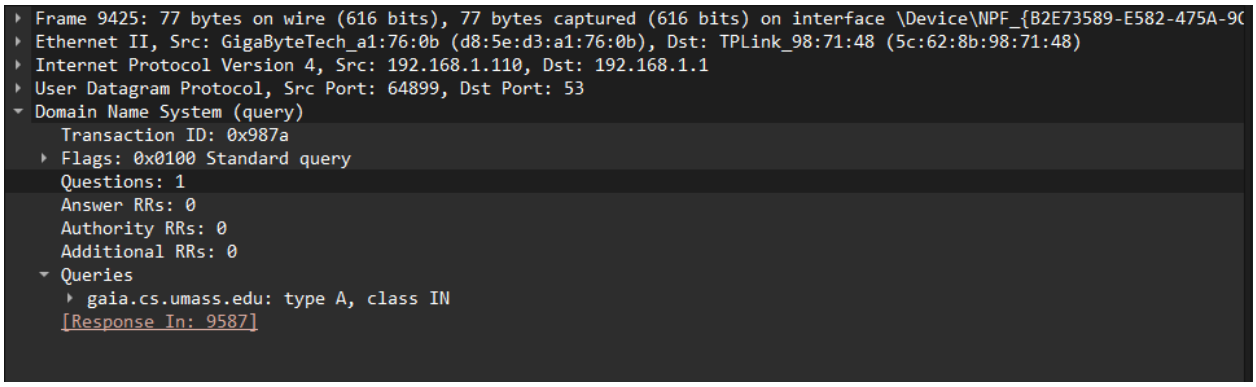
1) DNS Query ปกติจะมี Questions เป็น 1 คือขอข้อมูล A record ของ gaia.cs.umass.edu และ มี Answers เป็น 0 เพราะเป็นแค่คำขอ ไม่ใช่คำตอบ



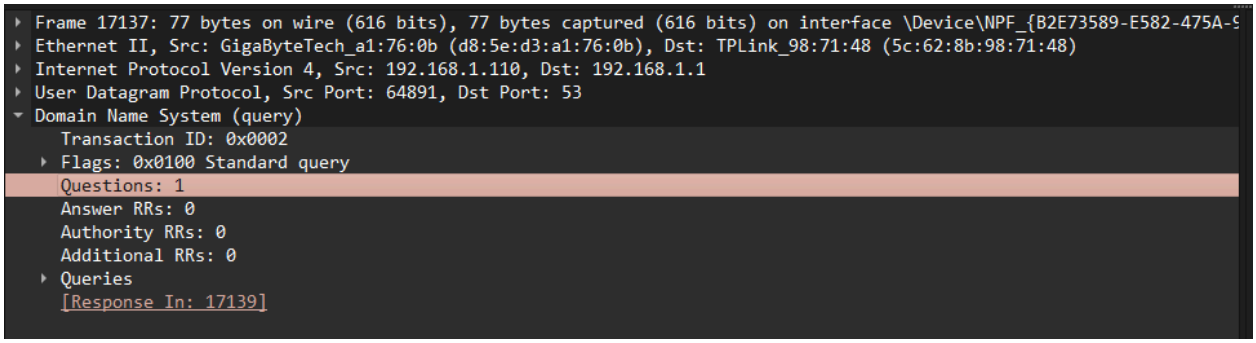
9420	20.579556	192.168.1.110	192.168.1.1	DNS	77 Standard query 0x2647 A gaia.cs.umass.edu
9421	20.579782	192.168.1.110	192.168.1.1	DNS	77 Standard query 0x372f HTTPS gaia.cs.umass.edu
9425	20.582346	192.168.1.110	192.168.1.1	DNS	77 Standard query 0x987a A gaia.cs.umass.edu
9426	20.582557	192.168.1.110	192.168.1.1	DNS	77 Standard query 0x8e50 HTTPS gaia.cs.umass.edu
17137	45.767480	192.168.1.110	192.168.1.1	DNS	77 Standard query 0x0002 A gaia.cs.umass.edu

```
Frame 9420: 77 bytes on wire (616 bits), 77 bytes captured (616 bits) on interface \Device\NPF_{B2E73589-E582-475A-9C
Ethernet II, Src: GigaByteTech_a1:76:0b (d8:5e:d3:a1:76:0b), Dst: TPLink_98:71:48 (5c:62:8b:98:71:48)
Internet Protocol Version 4, Src: 192.168.1.110, Dst: 192.168.1.1
User Datagram Protocol, Src Port: 51435, Dst Port: 53
Domain Name System (query)
    Transaction ID: 0x2647
  ▸ Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  ▾ Queries
    ▸ gaia.cs.umass.edu: type A, class IN
    [Response In: 9586]
```

2) DNS Query ปกติจะมี Questions เป็น 1 คือขอข้อมูล A record ของ gaia.cs.umass.edu และ มี Answers เป็น 0 เพราะเป็นแค่คำขอ ไม่ใช่คำตอบ

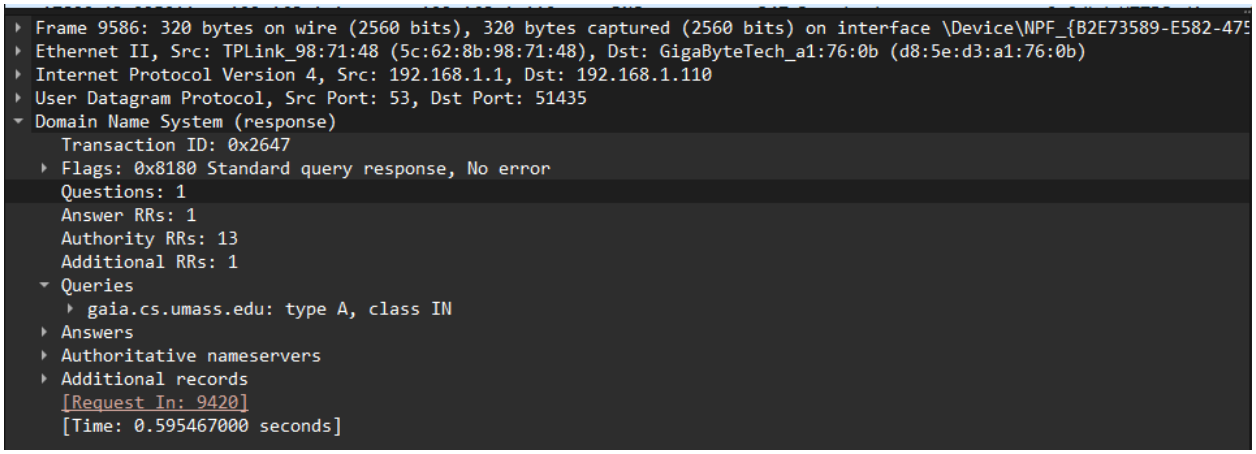


3) DNS Query ปกติจะมี Questions เป็น 1 คือขอข้อมูล A record ของ gaia.cs.umass.edu และ มี Answers เป็น 0 เพราะเป็นแค่คำขอ ไม่ใช่คำตอบ

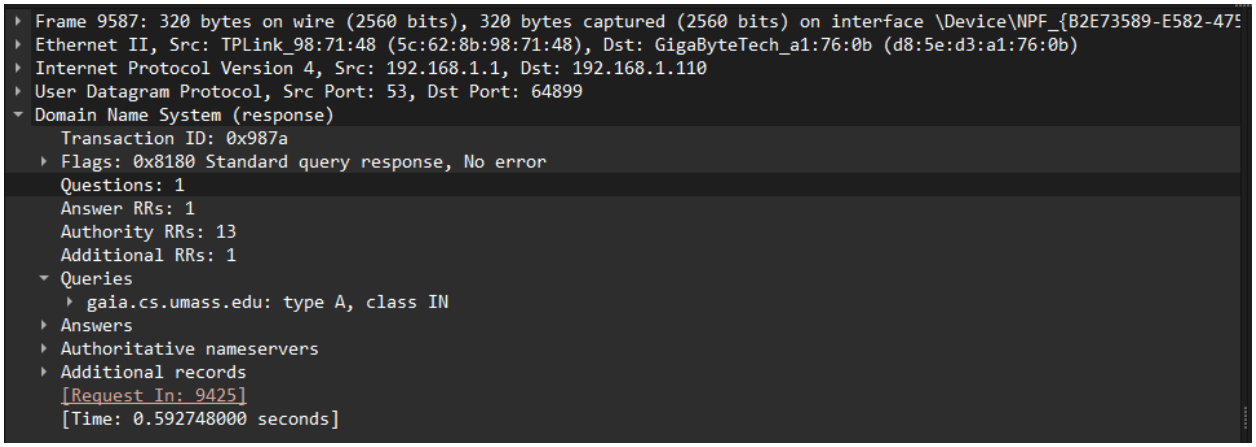


10. Examine the DNS response message to the initial query message. How many “questions” does this DNS message contain? How many “answers” answers does it contain?

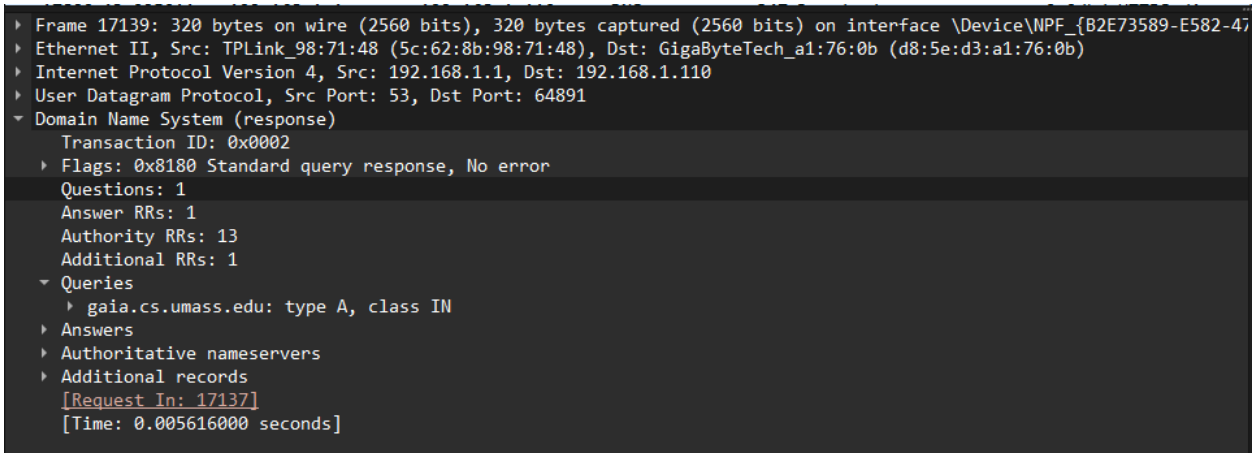
ANS



1) DNS Response มี Questions เป็น 1 และ Answers เป็น 1



2) DNS Response มี Questions เป็น 1 และ Answers เป็น 1



3) DNS Response มี Questions เป็น 1 และ Answers เป็น 1

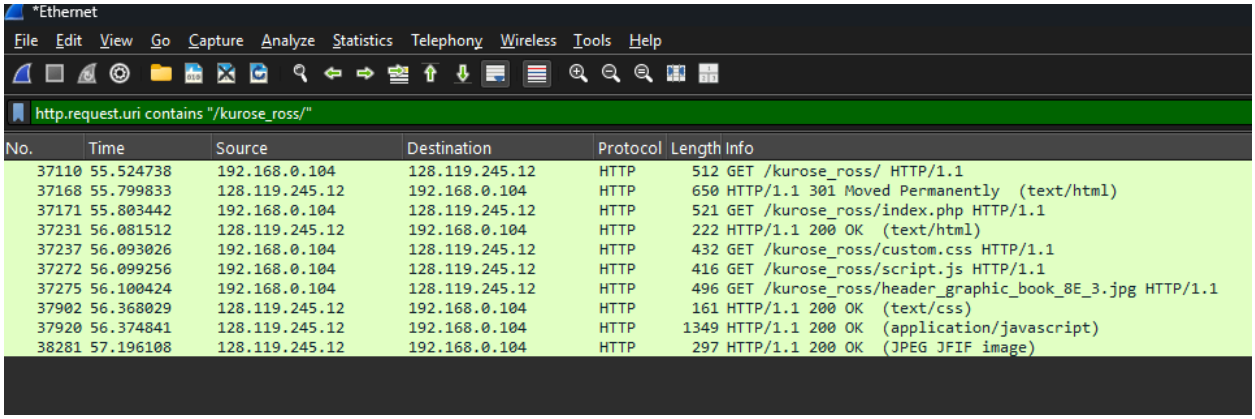
11. The web page for the base file http://gaia.cs.umass.edu/kurose_ross/ references

the image object

http://gaia.cs.umass.edu/kurose_ross/header_graphic_book_8E_2.jpg , which, like

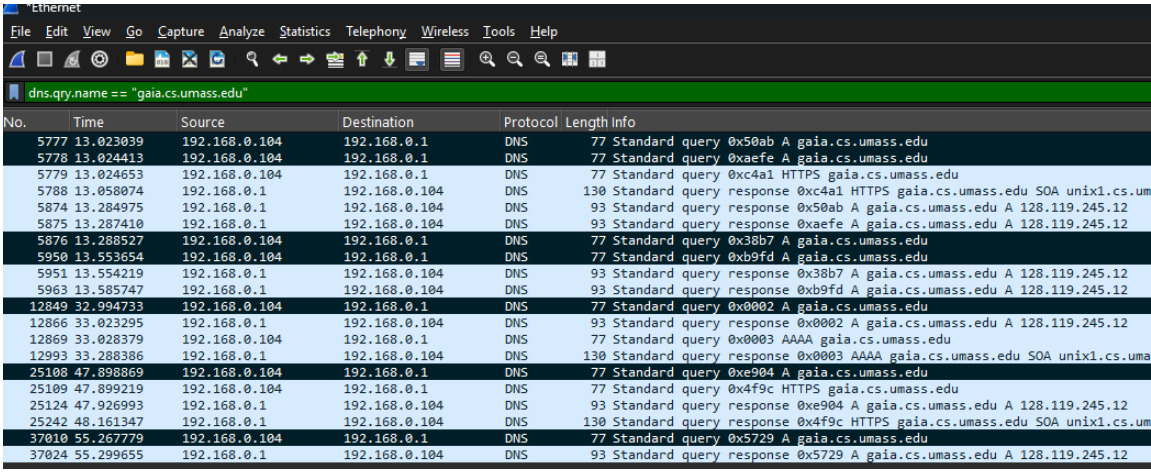
the base webpage, is on gaia.cs.umass.edu..

1) What is the packet number in the trace for the initial HTTP GET request for the base file http://gaia.cs.umass.edu/kurose_ross/?



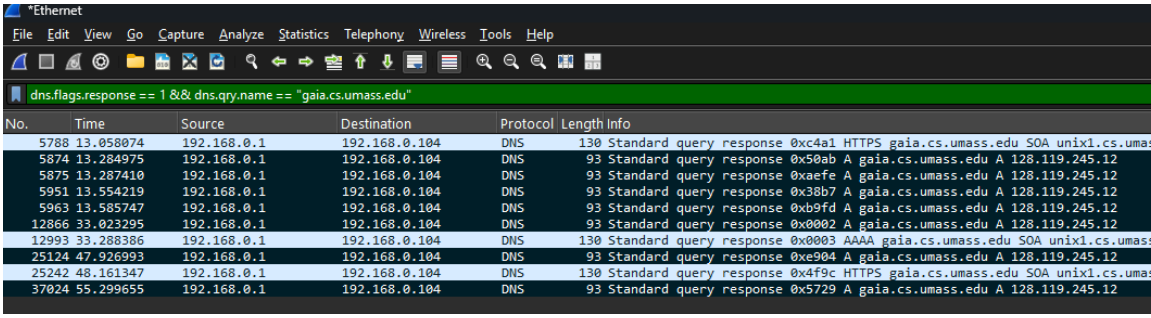
ANS 1.Packet #37110

2) What is the packet number in the trace for the initial HTTP GET request for the base file http://gaia.cs.umass.edu/kurose_ross/? What is the packet number in the trace of the DNS query made to resolve gaia.cs.umass.edu so that this initial HTTP request can be sent to the gaia.cs.umass.edu IP address?



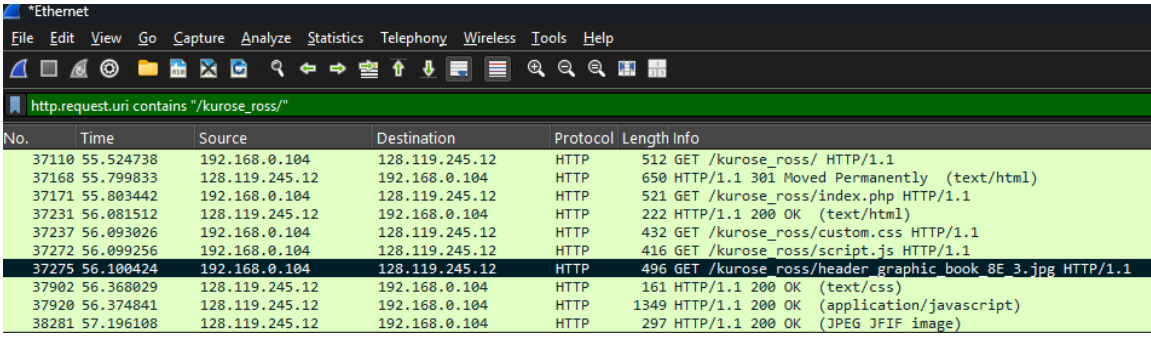
ANS No.markPacket

3) What is the packet number in the trace of the received DNS response?



ANS No.markPacket

4) What is the packet number in the trace for the HTTP GET request for the image object http://gaia.cs.umass.edu/kurose_ross/header_graphic_book_8E2.jpg?



5) What is the packet number in the DNS query made to resolve gaia.cs.umass.edu so that this second HTTP request can be sent to the gaia.cs.umass.edu IP address?

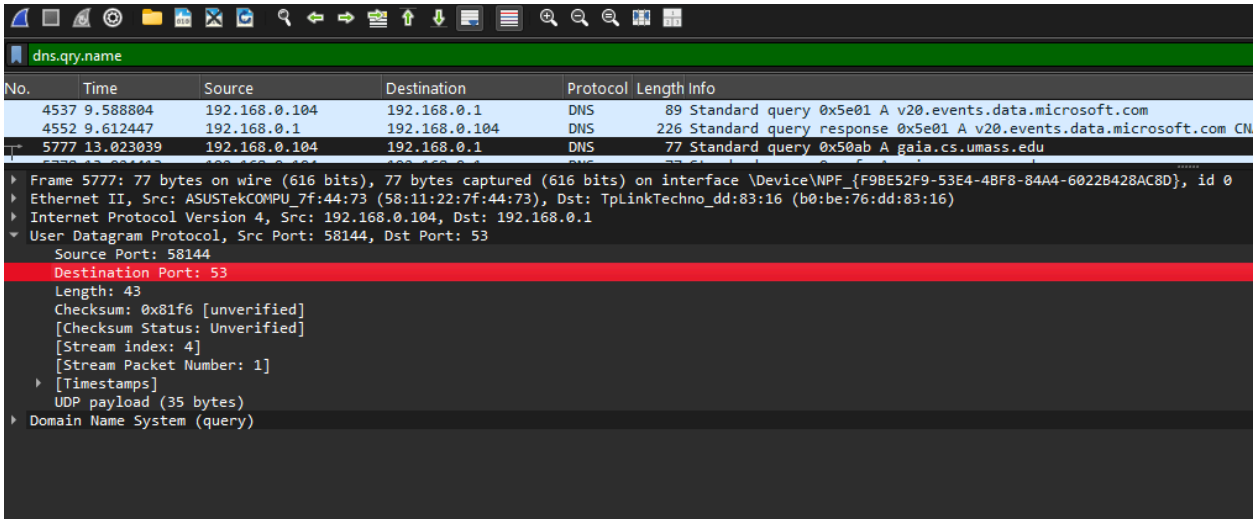
ANS ไม่มี DNS Cache ใช้งาน

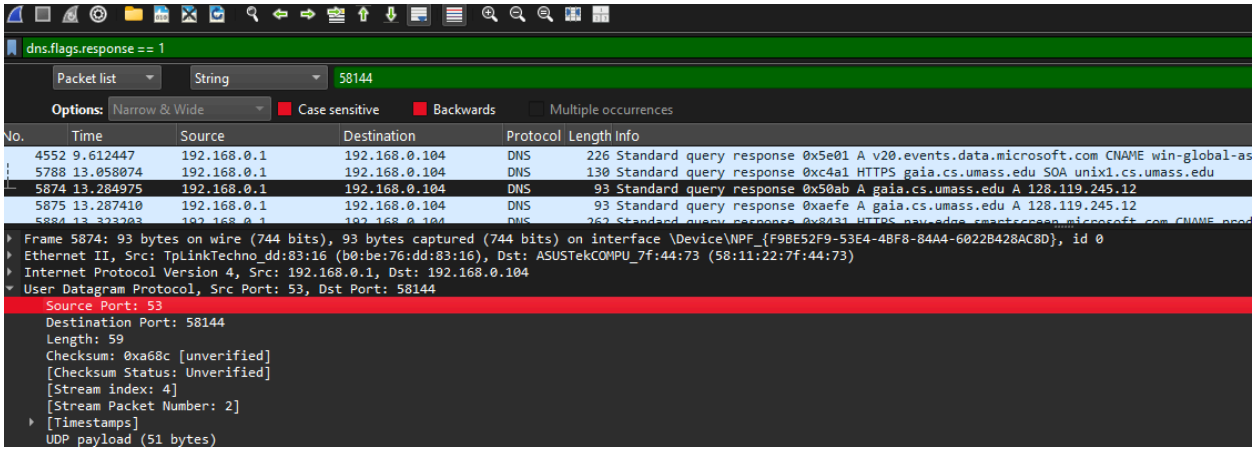
12. What is the destination port for the DNS query message? What is the source port of the DNS response message?

ANS

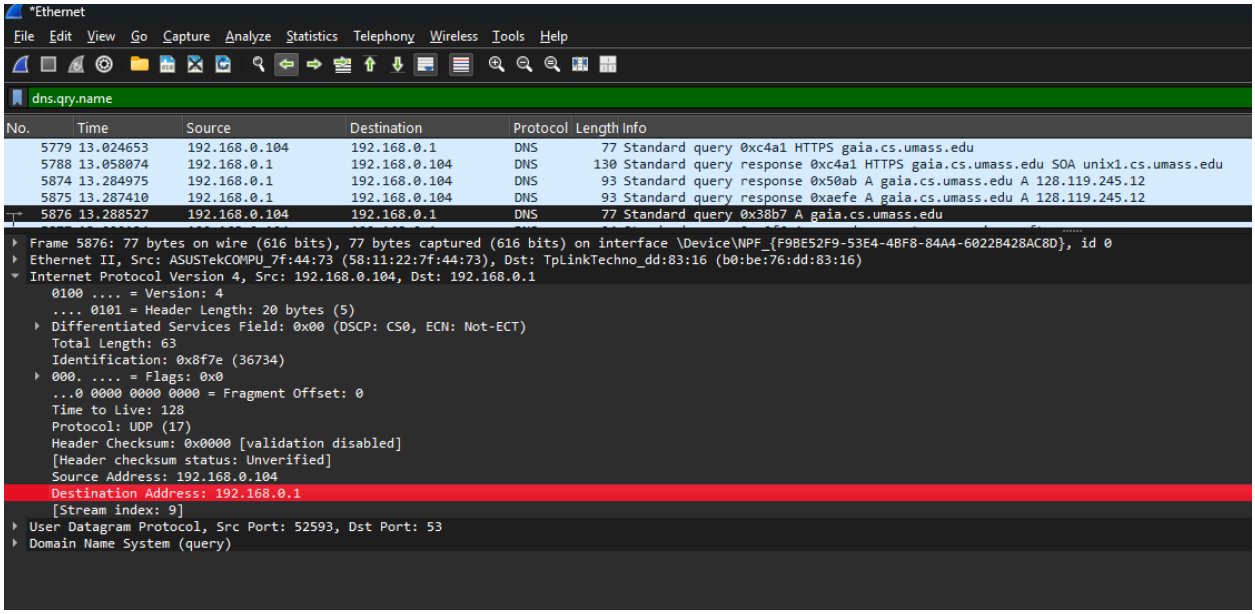
Destination Port คือ 53

Source Port คือ 53





13. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?

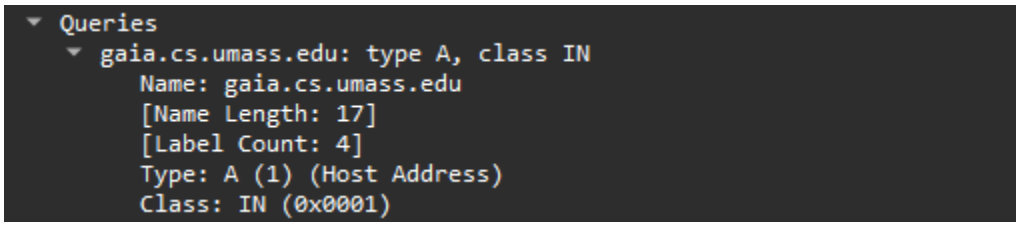


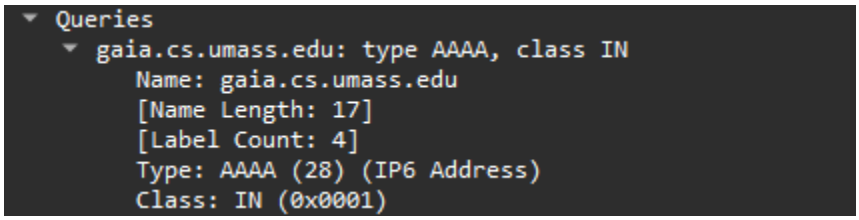
ANS

IP address ที่ส่งมา เป็น local default DNS server ที่ IP 192.168.0.1

14. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?

ANS

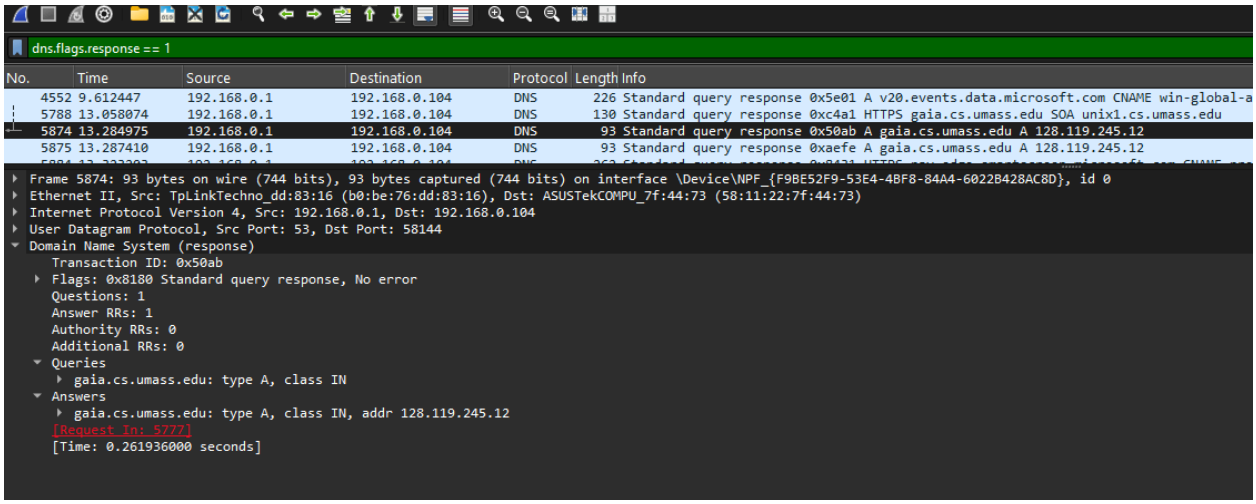




Query Type ของ DNS คือ A(IPv4)และ

QueryType ของ DNS คือ AAAA(IPv6)

15. Examine the DNS response message to the query message. How many “questions” does this DNS response message contain? How many “answers”?



ANS

query ส่งข้อความกลับมามี IP 1 Question และมี 1 Answer (128)

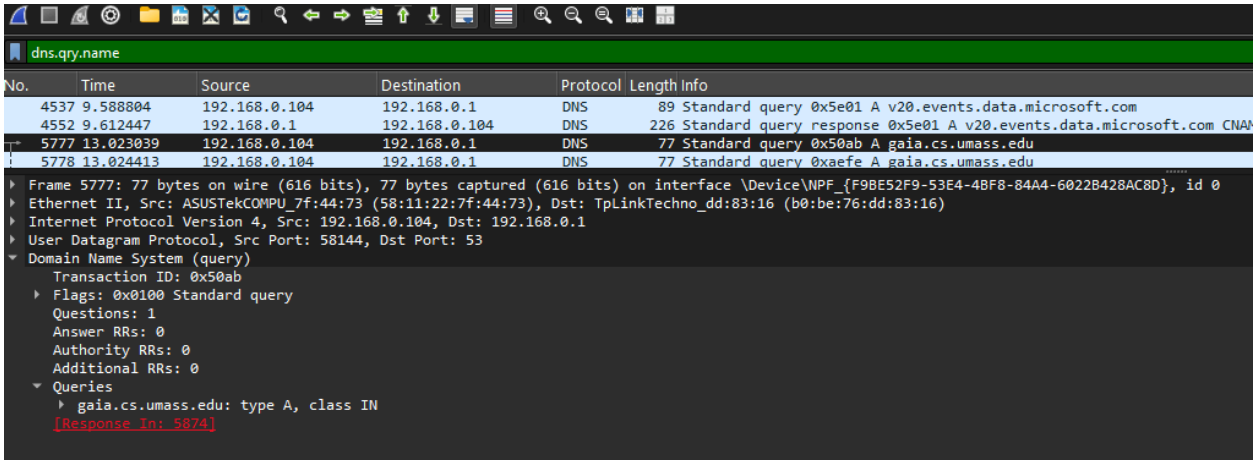
16. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?

ANS

IP address ที่ส่งมา เป็น local default DNS server

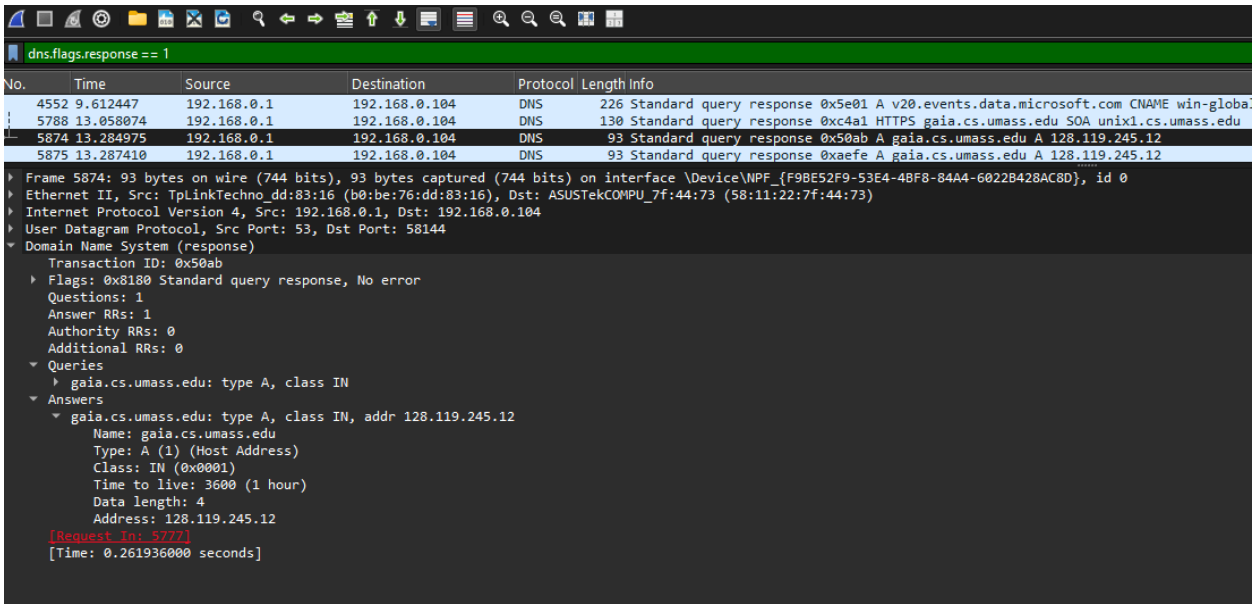
ที่ IP 192.168.0.1

17. Examine the DNS query message. How many questions does the query have? Does the query message contain any “answers”?



ANS query มี 1 Question และ ไม่มี Answer ไม่มี

18. Examine the DNS response message. How many answers does the response have? What information is contained in the answers? How many additional resource records are returned? What additional information is included in these additional resource records?



ANS มี 1 Answer ⟶ IP Address 128.119.245.12